

WiseWanderer App

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ABSTRACT

This study addresses the safety concerns of solo travelers, particularly women, by designing and evaluating a mobile application that prioritizes real-time assistance, safety insights, and community-based features. Recognizing a gap in travel apps that specifically cater to the needs of solo users, our project focuses on enhancing perceived and actual safety during travel experiences. The design includes features such as onboarding with verification, trusted contact sharing, destination safety search, buddy requests, real-time navigation with alerts, and an emergency SOS system.

To assess the usability and effectiveness of the app, we conducted structured usability testing with five participants aged 20–26. Each user completed a predefined set of tasks, including onboarding, safety-related searches, and emergency simulations. Key usability dimensions such as learnability, efficiency, memorability, error tolerance, and satisfaction were measured through ratings and qualitative feedback.

Findings indicate high overall satisfaction, with users appreciating features like easy onboarding, clear SOS functionality, and smooth trusted contact management. However, areas such as real-time navigation and filtering during searches revealed complexity and cognitive load, suggesting a need for interface simplification. The average usability ratings across dimensions ranged between 4.0 and 4.5, indicating a generally positive user experience.

The contribution of this work lies in presenting a user-centered solution that integrates safety and travel planning in a cohesive interface. Through iterative design and evaluation, this project demonstrates how digital tools can empower solo travelers by increasing their autonomy, awareness, and preparedness in unfamiliar environments.

KEYWORDS

Solo travel, Women’s safety, Usability testing, User experience, Mobile application design, Human-centered design, Safety technology

1 INTRODUCTION

Solo travel has become more popular with youth as a way of achieving independence, exploration, and self-enrichment. Independent travel is usually described as having serious safety concerns for women and minorities. There is little good local knowledge, little assistance in case of an emergency, and little capacity to assess the safety of unvisited territories that can make independent travel stressful or even dangerous. Although there are travel and navigation applications, few of them integrate into their design the specific needs of women traveling independently.

This project responds to the design challenge of enabling independent travel—particularly women—by offering a digital solution that

increases their sense of safety, trust, and readiness during travel. The most important user requirement is a system that not only facilitates functional tasks such as navigation or emergency communication but also provides confidence with trustworthy advice, sound local knowledge, and an understandable user experience.

The general aim of this project is to develop and validate a mobile application that supports individual travelers throughout their journey life cycle: planning and onboarding, real-time safety routing, and reflection after the journey. Through a combination of usability testing and user-centered design, the project will aim to identify pain points in current travel safety solutions and recommend features that bridge these gaps clearly.

The contributions of this paper are threefold:

- a detailed usability evaluation of key app features through user testing
- identification of core usability trends and user pain points
- actionable design recommendations based on user feedback to improve safety, usability, and overall experience for solo travelers.

2 LITERATURE REVIEW

2.1 Similar Systems and Tools

There have been various technological solutions put in place in the past couple of years to assist in keeping women safe while traveling. Safety-oriented mobile apps typically have features like SOS alerts, sending your location via GPS, geofencing, and reporting an incident. The apps assist in keeping people connected with their trusted contacts and getting immediate assistance when required. Smart wearable technology, such as smart jewelry with discreet panic buttons, provides users with silent means of calling for assistance and broadcasting their location.

Community-oriented platforms are becoming increasingly popular. They provide peer safety recommendations, local knowledge, and access to trusted female local guides. They enhance social support and allow women to feel safe through experience sharing. Tour guides and Internet of Things devices are also assisting groups with safe navigation and offering emergency functionality. AI navigation devices provide personalized route plans. They avoid high-risk or dark spots with real-time knowledge and crime reports.

Despite all of these advancements, most of the tools have still failed to provide psychological comfort, cultural sensitivity, and individual needs to solo women travelers. These inadequacies have directed our project’s growth, which pays equal attention to safety as well as user independence, emotional comfort, and confidence.

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2.2 Theoretical Foundations and Design Principles

Our design approach is informed by a range of theoretical models and usability heuristics to meet the individual and cumulative needs of solo women travelers.

Empowerment-Focused UX Design emphasizes confidence, autonomy, and good judgment through user interfaces. Research indicates that simple, intuitive interfaces empower users more than interfaces that support maximum customization.

We investigated the journey of individual women tourists, especially those from Asian countries, using **Constructivist Grounded Theory**. Self-agency, risk, and transformation were of interest. From this approach, we can see that traveling is not just passing from one location to another; it involves dealing with social encounters and personal change.

Schmitt's Experience Design Model helped us to divide the user experience into five elements—sense, feel, think, act, and relate. These elements guided us in designing emotional and mental involvement within the app.

Hofstede's Cultural Dimensions Theory described the ways in which cultural values of power distance and masculinity affect the desire and self-confidence of women to travel independently, particularly in more conservative societies.

Participatory Design was instrumental in finding real pain points in the real world. By engaging the users directly, we caught on to common issues that are typically ignored—such as interface clutter, confusing navigation, and the absence of rapid emergency help features.

2.3 Usability Findings and Design Implications

One of the most important things we noticed in our literature review was that users distinguish between aesthetic enjoyment and functional usefulness. While beauty was appreciated, usability—like good organization, rapid access to safety aids, and error recovery—was more critical to the experience.

All the studies highlighted simplicity of UI, minimizing cognitive load, and simplicity of safety features to access and use as being most critical. Emotional comfort and trust were key themes as well. Women were concerned with physical threats such as harassment or robbery, but also with digital threats (e.g., being stalked on the internet) and social judgment (e.g., being judged for traveling alone). Our strategy takes all of these factors into account together—supporting users in practical, emotional, and social ways. Through learning from past research, we aimed to build a system that not only improves physical safety but also gives solo women travelers the confidence, resources, and supportive community so that they can travel alone and safely.

3 METHODOLOGY

This section outlines the methodological framework and research practices adopted throughout the project, emphasizing a user-focused, iterative design process.

3.1 Design Approach

The project followed an **Agile design process**, which enabled us to have ongoing feedback and iteration. We iterated from developing the concept to wireframing, storyboarding, and creating a mid-fidelity prototype. Having created the prototype, we performed usability testing with five participants. Their feedback was incredibly helpful in streamlining the design, especially in making cognitively challenging interfaces like the real-time navigation feature easier to use.

Agile was favored over linear approaches like the Waterfall model because of its responsiveness and capacity to adapt to feedback from users. Unlike Waterfall, which has fixed design and testing phases, Agile allows for rapid iteration and continuous improvement based on real feedback and is thus most well-suited to user-driven digital product development.

While Agile was our development guiding process, overall our design philosophy is based on principles of **Human-Centered Design (HCD)**. HCD prioritizes hearing the user's voice in the form of direct, face-to-face discussion and iterative prototyping. Our two-stage user research—a survey of 50 participants and 6 interviews of solo women travelers—ensured that user voices were kept at the center of the design and test process. This combination of Agile and HCD allowed us to design a more responsive, engaging, and resilient solution that is appropriate for the target user.

3.2 Research Methods

The research approach to this project was a two-stage process identifying independent female travelers' needs and pain points, focusing on safety. Broad and deep approaches toward understanding user behavior, preferences, and expectations were carried out at these two stages.

3.2.1 Phase 1: Survey (50 Participants). The first research stage was a survey of 50 participants, gender-balanced and with an age range of 19 to 45 years. This was to gather general opinions on users' perceptions of safety and unsafety when traveling alone. It covered the use of safety equipment, technology, and general concerns regarding traveling alone.

Major findings of Phase 1 include:

- **Safety Concerns:** Harassment, unsafe transportation, theft, and fraud were concerns for participants. These concerns were of far more importance in unknown surroundings.
- **Safety Tools Usage:** Most respondents were not actively using safety apps, mainly due to a lack of awareness or because they were using ineffective tools that did not provide real-time aid.
- **Emergency Response:** Overall, there was dissatisfaction with late responses by emergency services, and most believed that real-time safety features were lacking in current solutions.
- **Accommodation Safety:** There was extremely low confidence in host verification practices, with specific concerns over the safety of women in mixed-sex accommodation.

3.2.2 Phase 2: Interviews (6 Users). After the survey, Phase 2 consisted of in-depth one-to-one interviews with six women. This phase aimed to gather more in-depth information regarding the

particular pain points and requirements of the users, with a focus on real-life situations, subtle safety issues, and technology usage. The interviews provided more individualized responses that were not feasible in the large-scale survey. **Phase 2 major findings include:**

- **Specific Safety Issues:** Along with the threats of theft and harassment, the interviewees also mentioned stalking, unwanted attention, and internet safety threats, which were less explicitly stated in the survey.
- **Emergency Response Concerns:** More particular complaints were noted by users regarding, for instance, slow police response or the absence of area assistance in emergency scenarios.
- **AI & Technology Integration:** There was interest in AI-based safety solutions, but there was no intention to rely completely on AI recommendations. Interviewees preferred a mixture of human verification and AI recommendations, meaning a balanced approach was required for safety.
- **Safety in Accommodation:** Most interview respondents were concerned with women's safety in co-gender accommodations, and the preference was for certified local guides, women-only areas, and government integrations for safer assurance.

3.2.3 User Journey Maps (UJMs). User journey maps were created to show the solo female traveler's experience at each stage of their journey, from initial awareness of safety risks to giving post-trip feedback. The journey maps were sketched out for a generic user and for each of the four personas, including key actions, touch-points, pain points, and insights. For example, in the *Planning & Booking* phase, everyone expressed the need for reliable safety information before booking travel. While some of the participants were highly technology-reliant, for example, GPS and safety apps, others used more traditional sources, such as word of mouth or government guidance. These were addressed in the persona-specific UJMs because they represented alternative priorities and requirements.

User Journey Map - Solo Female Traveler Safety						
Stage	Awareness	Consideration	Planning & Booking	Travel & Navigation	On-Trip Experience	Post Experience
User Actions	Thinks about safety risks while planning travel.	Researches destinations, checks safety ratings & tips.	Books accommodation, transport, and safety tools.	Uses navigation apps, stays alert to surroundings.	Engages in activities while monitoring safety.	Shares experiences, safety feedback online.
Thoughts & Feelings	"Is this destination safe for solo travel?"	"I need reliable safety info before making decisions."	"I hope my accommodation and transport are secure."	"Am I being followed? Is this cab safe?"	"I want to enjoy, but I must stay cautious."	"Should I warn others about this place?"
Touch Points	Travel forums, social media, word-of-mouth.	Safety apps, government advisories, reviews.	Booking platforms, safety tool apps, GPS tracking.	Maps, ride-sharing apps, emergency contacts.	Local guides, buddy systems, safety check-ins.	Travel groups, safety rating platforms.
Pain Points	Lack of reliable, real-time safety information.	AI safety ratings feel generic, not tailored.	Booking sites lack verified security details.	Unreliable public transport & cab safety.	Safety concerns in public spaces, scams.	No follow-up from safety platforms.

3.2.4 Task Analysis for Usability Testing. To ensure the app was accessible to its target users, we conducted usability testing with five participants. The usability testing tasks were drawn from the app's

primary use cases, which were established through the process of user research. The tasks were:

- Sign up and complete the onboarding process.
- Add/manage a trusted contact and set sharing permissions.
- Complete a trip and leave feedback.
- Search a destination and check safety.
- Request a buddy / local guide.
- Use the real-time navigation feature.
- Use the emergency SOS feature.

The usability tasks were designed to mimic the most significant interactions that the users would make in the application. The tasks were also chosen with utmost attention to capturing any usability issues, especially in regards to cognitive load, efficiency, and error frequency. The feedback from usability testing helped to improve the interface, which became simpler and more intuitive, with reduced cognitive load, especially for tasks like real-time navigation.

4 DESIGN PROCESS

Our design process followed an iterative, user-centered approach rooted in the insights gathered during the research phase. The goal was to build a solution that aligned with real user needs while remaining minimalist, efficient, and accessible. The process was broken down into three key stages: problem framing & user research, ideation & concept development, and finally sketching, wireframing, and prototyping.

4.1 Problem Framing & User Research

To get a better understanding of the issues of solo female travelers, we conducted a two-phase user study. The first was the survey of 50 users, who were gender-balanced and ranged between 19 and 45 years. This assisted us in gathering general information on the use of technology, attitudes, and behaviors toward safety. We found from the survey that many participants encountered harassment, dangerous transportation, scams, and low trust in emergency systems and accommodation safety. Many users were not aware of available safety apps or found them useless since they lacked real-time support. To provide added depth to these findings with more context-rich insights, the second stage involved one-to-one interviews with six women. These revealed more specific issues of pain such as stalking, unwanted attention, and online safety concerns. We also discovered distrust in AI-only safety features, with users requiring a mix of human-vetted features, government integrations, and community-led resources. Drawing from the two-layered solution, we defined our problem as needing to create a simple, easy-to-use app for individual travelers, especially women, that thoroughly covers safety concerns at different stages of a travel. The solution had to reduce cognitive load, provide only necessary but useful features, and facilitate ease of use at different levels of technological savvy.

4.2 Core Use Cases Derived from User Research

Once we synthesized the user research findings, we moved into the ideation phase. Our aim was to define a set of core features—derived directly from user pain points and desires—that would form the backbone of our app.

After several rounds of brainstorming and feature prioritization, we identified the following seven key use cases:

- (1) Sign up and onboarding
- (2) Add/manage trusted contacts and set sharing permissions
- (3) Search a destination and check safety
- (4) Request a buddy/local guide
- (5) Real-time navigation
- (6) Emergency SOS
- (7) Post-trip feedback

These use cases were chosen for their criticality to user safety and their practical value in real-world scenarios. Additionally, we found that many of these features were missing or poorly implemented in existing market alternatives. We positioned our solution to fill these gaps while maintaining a seamless and intuitive user experience.

4.3 Sketching and Wireframing

Before diving into digital design tools, we initiated the process with hand-drawn sketches on paper. These early sketches were critical for rapidly visualizing ideas and aligning on the user flow across different features. The informal nature of sketching helped in iterating through layout possibilities without the cognitive overhead of detailed design elements.

After finalizing the basic flow and content arrangement, we transitioned to low-fidelity wireframing in Figma. These wireframes were created using the iPhone 14 Plus frame size to simulate a realistic mobile interface. Following principles of lo-fi design, we kept elements minimal—using placeholder text (*lorem ipsum*), gray-scale color palettes, boxes to represent buttons or images, and standard icons for navigational cues.

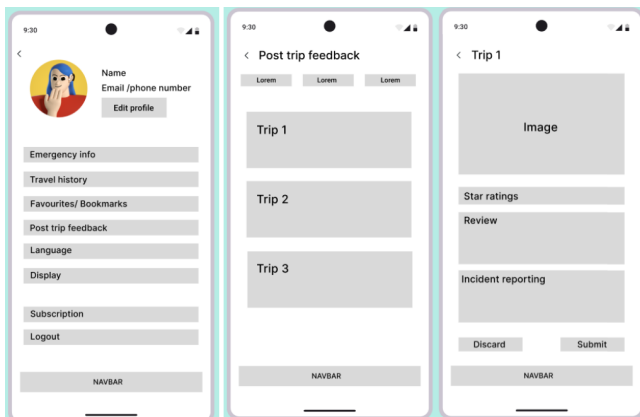


Figure 1: Sample Wireframe of Post trip feedback use case

The goal of wireframing was to:

- Establish the screen hierarchy and logical progression between tasks.
- Validate the usability and efficiency of navigation structures.
- Minimize distractions and focus solely on structure, spacing, and user journey.

This stage was instrumental in refining the layout based on internal feedback and served as a stepping stone toward higher fidelity development.

4.4 Storyboarding and High-Fidelity Prototyping

Once the wireframes were done and internal consensus was achieved, we moved on to high-fidelity prototyping, again in Figma. Before creating the interactive prototype, we made storyboards for each use case beforehand. A storyboard allowed us to describe the step-by-step process a user would follow, specifying screens, user actions, and scenarios. This helped visualize and plan our users' functional and emotional journey.

We then applied these storyboards to create high-fidelity screens with a uniform visual design of orange and white across the app. We employed orange to indicate alertness and warmth, and white for openness and transparency. For urgent situations such as SOS or emergency messages, we used a black and red color scheme to convey seriousness and urgency, offering high visual contrast to reduce response time.

Other design choices included:

- Clean typography for readability.
- High contrast between items for accessibility.
- Limited on-screen options to reduce decision fatigue and cognitive load.

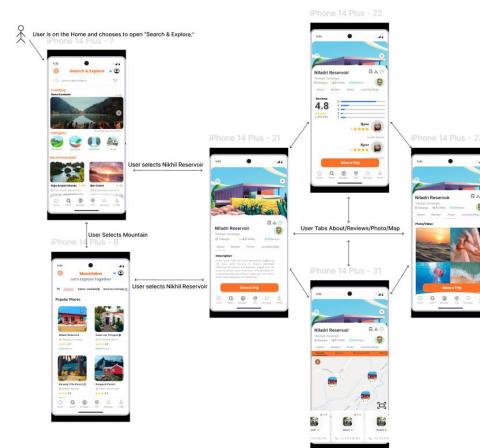


Figure 2: Storyboard for “Explore & Search” use case

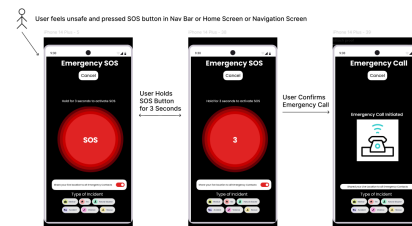


Figure 3: Storyboard for “Emergency SOS” use case

Using Figma's prototyping functionality, we connected every screen to logical navigation routes, simulating actual interactions such as button taps, transitions, and screen swipes. The prototype was interactive and usability test-ready, enabling realistic task completion and feedback gathering.

5 SYSTEM DESCRIPTION

5.1 Navigation Model

The proposed mobile application follows a **Hub-and-Spoke navigation model**, where a central hub (typically the home screen or dashboard) connects users to multiple independent modules such as Travel Planning, Explore, Navigation, Trusted Contacts, and Profile. This model is particularly well-suited for safety-critical apps, as it provides a predictable and clear structure that minimizes user confusion, especially during emergencies. Each feature can be accessed directly from the hub without unnecessary depth or complexity, enabling quick transitions and reducing cognitive load. In the context of solo travel, where fast access to emergency features, trusted contact sharing, and verified information is essential, the Hub-and-Spoke model ensures that users are always just one tap away from critical functionality. Its simplicity and reliability make it a perfect navigation framework for building user trust, enhancing usability, and ensuring a secure travel experience.

5.2 Theme & Aesthetics

The final user interface design of the app was simple and minimalist, designed to reduce users' cognitive load in potentially stressful travel situations. The general visual aesthetic used warm, welcoming orange and white color scheme across the interface, giving an atmosphere of friendliness and vibrancy without becoming overwhelming to the user. The orange accents were used primarily on buttons, highlights, and active states, which made key actions stand out without having a visually disorienting background in white or light colors.

For critical interactions, specifically the Emergency SOS screen, a red and black theme was employed. The color shift had two impacts: it signaled urgency, and it made emergency actions graphically distinct and readily locatable. This use of contrast and color coding is in accord with cognitive design principles to make salient and readily accessible at a glance critical safety functions.

Typography was in a clean and modern Roboto, which is very legible on different screen sizes. Font weights were varied to distinguish headers, labels, and body text for improved scannability and reduced reading effort.

The UI components were created with a consistent layout grid, generous spacing, and big touch targets in order to make the product more accessible to on-the-go mobile users. Icons were clear and simple, used in combination with text to double-check meaning—especially for navigation, safety features, and trusted contact interactions.

Overall, the interface is a blend of beauty and simplicity, so that individual female travelers can effortlessly utilize the app, even in high-pressure or high-speed conditions.

5.3 Annotated Prototype

To clearly communicate the functionality and navigation flow of the solo travel safety app, an annotated prototype was created using Figma. This prototype visually illustrates how different screens and interface elements are interconnected, enabling smooth transitions between core features such as onboarding, trusted contacts, emergency SOS, and real-time navigation. Each annotation highlights key user actions and system responses, ensuring clarity for developers, stakeholders, and testers. The image below demonstrates the screen flow and interactions designed to support a seamless user experience.

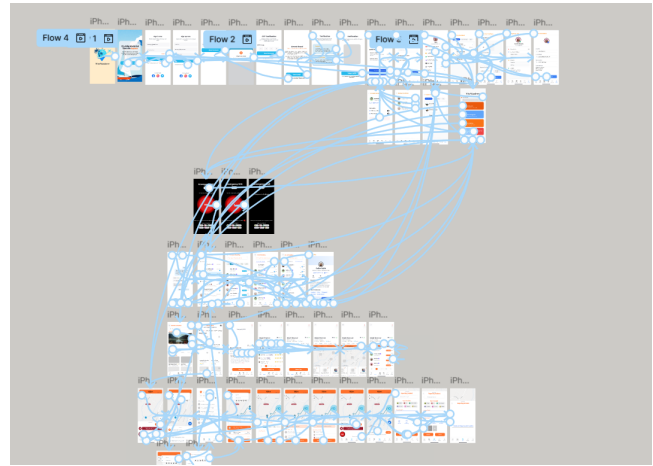


Figure 4: Visuals of how elements throughout the screens are connected for Navigation

5.4 Key features

The app integrates several thoughtfully designed features to address the specific safety needs of solo female travelers. Among these, the following stand out as unique:

- **Crowdsourced Safety Ratings for Locations:** Unlike typical navigation apps, this feature allows users to view real-time, community-generated safety scores for neighborhoods and travel spots. This participatory model provides local context and enhances user trust in decision-making.
- **Buddy System with Verified Local Guides:** Users can connect with pre-verified local companions or fellow solo travelers. This feature offers not only a sense of community but also human support during travels, bridging technology with trusted social interaction.
- **Multi-Modal Emergency Support:** The SOS functionality triggers alerts to emergency contacts, shares live location, notifies nearby trusted users, and provides step-by-step safety actions. This layered approach ensures faster and more effective responses in emergencies.

These features were developed through an iterative, user-centered design process and address key pain points identified during research, offering a holistic, empowering safety tool for solo travelers.

6 USABILITY EVALUATION

6.1 Evaluation Method

The usability evaluation for the app was conducted through a series of usability testing sessions involving five participants. These sessions aimed to assess the app's overall user experience, identify usability issues, and gather feedback to improve the design.

The evaluation utilized the think-aloud protocol, which required participants to verbalize their thoughts while interacting with the prototype. This method allowed us to observe the participants' cognitive processes, pinpoint areas of confusion or frustration during task completion, and gain valuable insights into the app's usability. By encouraging users to express their thought processes, we were able to understand their decision-making and identify barriers that hindered their flow.

The testing sessions focused on six primary tasks based on the app's core use cases:

- (1) **Sign Up and Onboarding:** Testing how easily users could create an account and navigate the setup.
- (2) **Add/Manage Trusted Contacts:** Evaluating the user's ability to add and configure trusted contacts and set sharing permissions.
- (3) **Complete a Trip and Leave Feedback:** Observing how users could complete a trip and leave feedback on their experiences.
- (4) **Search a Destination and Check Safety:** Testing the safety-check feature for specific destinations.
- (5) **Request a Buddy/Local Guide:** Evaluating the ease of booking a buddy or local guide.
- (6) **Real-time Navigation:** Assessing the real-time navigation functionality in practical settings.
- (7) **Emergency SOS:** Ensuring users could easily trigger an SOS and access emergency assistance when needed.

Each participant was observed during the test, asked to think aloud as they navigated the app, and then asked to provide feedback on each task, including ease of use, areas of confusion, and overall satisfaction with the experience. This is what the feedback matrix of each user looked like:

User	Task Description	Task Outcome	Time Taken	Main Pain Points	Things they liked	Task Rating (1-5)	Suggestions
U1	Sign up and complete onboarding	Completed	2 min	-	Verification is a nice step	4.5	Relevant info to be in bold
U1	Add/manage a trusted contact and set sharing permissions	Completed	1 min	-	easy to do	4	-
U1	Complete a trip and leave feedback	Completed	3 min	-	easy	4	-
U1	Search a destination and check safety	Completed	2-3 min	No option to sort acc to factors?	-	3.5	Uses category option and budget option, - connect w google maps
U1	Request a buddy / local guide	Completed	4min	-	Good that I can see one building, enter profile before sending request	4	Encryption
U1	Real time navigation	Completed	3-4min	very complex st	-	3	make the page a bit easier to comprehend
U1	Emergency SOS	Completed	<1 min	-	-	5	-

Figure 5: Anonymous, 20 yr old, Female's feedback of Usability Testing

6.2 Findings

6.2.1 Qualitative Results. Qualitative findings from usability testing exposed several insights into the user experience with the app. Users tended to value the clearness and simplicity of the **Emergency SOS** and **Real-time Navigation** features, which were both commended for ease of use and reassuring design. Users believed that these features gave a sense of safety that was very good, especially for lone travelers.

Nonetheless, activities like **Add/Manage Trusted Contacts** and **Request a Buddy/Local Guide** were challenging because instructions were not clear and there was no guidance, and some users became confused. More personalized options like custom notifications or safety update preferences were also recommended by feedback. Users also requested adding clearer labels and tooltips to these advanced features to enhance their overall usability.

Furthermore, although the **Onboarding Process** as a whole was widely appreciated for its ease, users indicated they wanted more specific guidance when setting up certain features of the app.

In spite of these issues, participants generally felt content with the app and commented that it fulfilled their primary safety requirements during travel. The key finding from these qualitative results was that although the app excels in critical areas, additional refinements to guidance and personalization would further improve the user experience.

6.2.2 Quantitative Results. Quantitative findings from usability testing were collected via structured task ratings on a 1 to 5 scale with five participants, addressing all major use cases of the app. The **Usability Matrix** showed excellent performance in major areas such as **Learnability (4.5)** and **Satisfaction (4.3)**, reflecting that users found the app easy to use and were overall satisfied with their experience. **Efficiency** also rated highly at **4.2**, indicating that users were completing most tasks quickly with little friction.

However, lower scores in **Memorability (4.0)** and **Errors (4.1)** suggested some usability problems on occasion, especially in multi-step flows such as **Requesting a Buddy** or **Using the Navigation/Search Features**. These scores were computed by correlating task performance (such as success rate, time on task, and ease rating) to each usability metric and averaging them for all users.

Although there were some shortcomings here and there, the overall metrics indicated that the product is functionally sound, easy to use, and perceived as being useful and reliable in actual usage situations. Numerical outcomes validated the general conclusion that the app is designed well, albeit with opportunity for optimization where there are higher levels of complexity in interaction paths.

6.2.3 Key Usability Successes.

- **Smooth and Quick Onboarding:** Users found the sign-up and verification process fast and intuitive. Features like selfie capture added to a seamless flow.
- **Trusted Contacts and Feedback Features Work Well:** Adding and managing trusted contacts was straightforward. Post-trip feedback was quick and easy, though a few suggested reminders could enhance the experience.
- **Emergency SOS is Quick and Effective:** The SOS feature was highly appreciated for its speed and clarity. It was seen as efficient and essential for solo travelers.

- **Local Guide Profiles Build Trust:** Users liked viewing detailed and verified local guide profiles before making requests, which built credibility and confidence.

6.2.4 Key Usability Issues.

- **Real-Time Navigation Needs Better UI:** Several users found the navigation interface cluttered or visually harsh. Suggestions included increased spacing, voice guidance, and a cleaner layout.
- **Lack of Filters and Sorting Options:** Users wanted more filters (e.g., safety level, language, location) and better sorting for destination and buddy features.
- **Design and Accessibility Concerns:** Small buttons (especially in the SOS section), tight card spacing, and font contrast were raised as accessibility and readability issues.
- **Minor Navigation and Prompting Issues:** Issues like non-intuitive back buttons and missing prompts (e.g., reminders to add emergency info or leave feedback) affected smooth flow for some users.

6.2.5 Usability Matrix. To systematically assess the effectiveness and user-friendliness of the application, a Usability Matrix was constructed. This matrix quantifies user feedback across five core usability dimensions—Learnability, Efficiency, Memorability, Error Rate, and Satisfaction—each rated on a scale from 1 to 5. The evaluation was based on direct user feedback obtained during usability testing sessions, where five participants performed representative tasks corresponding to each use case of the application. The individual scores were derived by mapping these usability factors to the relevant app tasks and computing the mean user ratings:

- **Learnability (4.5):** Primarily assessed through the "Sign-up and Onboarding" task, which was found to be simple and intuitive by most users.
- **Efficiency (4.2):** Evaluated using tasks such as Trusted Contact addition, Destination Search, and Buddy Request. These were completed quickly by participants with minimal friction.
- **Memorability (4.0):** Measured through repeated-use features like Post-Trip Feedback and Trusted Contact modification. A few users required brief reminders, slightly lowering this score.
- **Errors (4.1):** Focused on tasks prone to confusion, such as Real-Time Navigation and Search. While the overall error frequency was low, a few participants did encounter minor UI issues.
- **Satisfaction (4.3):** Derived from post-session surveys where users rated their overall experience. Most users responded positively, citing visual clarity and easy navigation as key strengths.

This matrix served as a structured benchmark to guide further UI refinements, reduce cognitive load, and enhance overall usability.

7 DISCUSSION

The solo female traveler safety app's design and development were informed by rigorous user research and iterative testing. This section outlines the major design decisions, their consequences, design trade-offs, limitations, and any surprising findings that were experienced throughout the project.

7.1 Design Decision Reflections

The major design decision was to implement a minimalist, easy-to-use interface that offered valuable safety features without inundating users with convoluted options. This choice was guided by the necessity of rapid access to safety equipment in high-pressure situations. Wireframes were deliberately made simple and clean, with an emphasis on intuitive interactions that reflected the needs and feedback of users in Phase 1 (survey) and Phase 2 (interviews). In addition, we put first-use safety features like real-time navigation, emergency SOS, and trusted contacts, prioritizing that the app could fulfill the solo traveler's immediate needs without clutter.

7.2 Implications for Users, Systems, or Society

This app's design has strong implications for the users and society as well. For solo women travelers, this app may act as an essential resource for mitigating perceived and real risks when they travel. By combining functionality such as real-time navigation with safety alerts, emergency SOS, and authenticated local guides, the app might be able to give users peace of mind and a feeling of control, particularly when traveling in new or dangerous places. On a larger scale, these solutions can help make travel safer and more accessible for women, promote greater independence, and inspire more women to take advantage of travel options that might otherwise have seemed too risky.

From a systems viewpoint, the creation of this app may have larger implications for the safety tech industry as a whole. It incorporates technologies such as AI-driven safety suggestions, GPS monitoring, and real-time alerts, joining the trend towards employing digital tools to solve safety problems. But with its hybrid system of human authentication and technology, it provides an innovative way of overcoming the risks of solely automated solutions, which may be open to errors or bias.

7.3 Design Trade-offs and Constraints

There were a number of design trade-offs and constraints that influenced the ultimate design. For example, though minimalism was emphasized in the user interface, this meant that the quantity of information presented at any one time was restricted. The balancing of the need for extensive safety information against the minimalist nature of the app's interface involved close attention to the most important elements to present. Moreover, certain users had a desire for more sophisticated features, i.e., AI-powered suggestions, but this was weighed against the aim of offering an easy-to-use and simple experience to everyone.

Another limitation was the possibility of battery life and data usage. Functions such as real-time GPS tracking and safety alerts could exhaust a phone's battery, so we needed to optimize these functions so that they would be useful without bogging down the device's resources. Also, privacy issues were an important factor, particularly

Usability Factor	Description	Avg. Rating (1–5)
Learnability	How easy is it for users to learn to use the product?	4.5
Efficiency	How quickly can users accomplish tasks with the product?	4.2
Memorability	How well can users remember how to use the product after a period of not using it?	4.0
Errors	How often do users make errors while using the product?	4.1
Satisfaction	How satisfied are users with their overall experience of using the product?	4.3

Table 1: Usability Matrix Summary

in functions such as location sharing, and we needed to make sure that users were in full control of their data.

7.4 Surprising Outcomes

Through the usability testing process, one of the most surprising outcomes was the high degree of variation in user preferences regarding emergency response. Although the majority of users anticipated real-time intervention via the app, a number of users highlighted the need for an immediate, face-to-face response (i.e., from neighborhood authorities or known contacts) in addition to pure app-oriented solutions. This observation refuted our initial hypothesis that customers would want a purely digital experience and underscored the demand for a blend of both digital systems and hands-on support.

Also, feedback from users indicated an unintended worry regarding digital safety, particularly regarding data privacy and how their location and travel information was being transmitted. This prompted the inclusion of stronger privacy controls within the app, allowing users to determine who could access their information and for what period of time.

8 LIMITATIONS & ETHICAL CONSTRAINTS

8.1 Limitations

The development and design of the solo female traveler safety app, though grounded in substantial user research, have some shortcomings. Among assumptions made while developing the project is that the users would experience steady internet access, particularly for aspects such as real-time location tracking and sending out emergency notifications. In actual application, however, internet may not be steady or even exist in some areas, which can hugely affect the operation of the app. Furthermore, testing of the application was carried out in a controlled setting with a small sample of five users under usability testing that might not necessarily reflect the variety of solo travelers. The gathered feedback can also be skewed based on the individual needs of the small group of users compared to the entire group of travelers. In addition, the project scope was only for creating a mobile app, without examining the possibilities of integration with other safety systems, including emergency response networks or local law enforcement. Lastly, the project did not fully test the app in different environmental settings or real-world situations, where user action and interaction with the app might be circumstantially influenced or stressed, potentially impacting the effectiveness of the app.

8.2 Ethical Constraints

The app also introduces a number of ethical concerns. The most notable of these relates to the data gathering, storage, and sharing of sensitive individual information, including location, emergency contact information, and travel itineraries. In order to overcome this, users must be offered informed consent prior to app use, allowing them complete clarity as to what data will be gathered and for what purposes. Security measures to protect data have to be in place to avoid illegal access to personal data. Also, since the app would depend on real-time sharing of and tracking of location information, privacy issues arise since people will not necessarily want to share their exact location with other people all the time. Another ethical concern is the possibility of bias in safety advice from algorithms, which might inadvertently prefer some areas or categories of safety hazards over others. The AI functionality of the app must be trained on representative datasets so that all users are treated equally. It is important to repeatedly check the ethical implications of such functions and make the necessary corrections to promote fairness and transparency.

9 CONCLUSION

In summary, solo female traveler safety app design and development fills a key gap in the market through user-centric solutions to safety issues. Through user research, we were able to determine key areas of pain and design an app that is functional, minimalist, and easy to use, with features specifically designed to improve the security and comfort of solo female travelers. The usability test exhibited the app’s ability to enhance user experience through high learnability, efficiency, and satisfaction scores. There is, however, room for development in real-world testing and more extensive integration into local emergency infrastructure. Future updates can include support for a diverse population of users, integration into outside safety networks, and such features as predictive warning of impending danger. Additional studies might also investigate the long-term efficacy of the app in different travel conditions and its effect on the general safety of solo travelers.

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A USER RESEARCH SUMMARY: SOLO WOMEN TRAVEL SAFETY

A.1 Phase 1 Survey

Questions:

- (1) What is your age group?
- (2) What is your gender?
- (3) How often do you travel alone?
- (4) What are your primary reasons for solo travel? (Select all that apply)
- (5) Which types of destinations do you usually visit?
- (6) What does the word 'safe' mean to you when you are travelling solo or in a group?
- (7) Have you ever felt unsafe while traveling alone?
- (8) What safety concerns do you face while traveling alone? (Select all that apply)
- (9) What safety measures do you currently take while traveling solo? (Select all that apply)
- (10) Do you currently use any digital safety tools while traveling?
- (11) What features do you find most useful in a travel safety app?
- (12) How important is community support for your solo travel safety?
- (13) Would you trust real-time, user-generated safety ratings for locations?
- (14) Would you use a platform that connects solo travelers with verified local guides or buddy travelers?
- (15) What features would you like to see in an ideal travel safety platform?
- (16) How would you prefer to receive safety alerts while traveling?
- (17) Would you be willing to pay for a premium version of a travel safety app with advanced security features?
- (18) What improvements do you think should be made in existing travel safety solutions?

A.2 Phase 2 Interview

Questions:

- **Section 1: User Demographics & Travel Persona**
 - What age group do you belong to?
 - How frequently do you travel alone, and what factors influence this?
 - What motivates you to travel solo?

- Can you describe the types of destinations you prefer to visit and why?

- **Section 2: Safety Concerns & Challenges in Solo Travel**

- Can you share an experience where you felt unsafe while traveling alone?
- What are the most common safety concerns you consider before traveling alone?
- How do you assess the safety of accommodations and transport before choosing them?
- How do digital tools influence your sense of safety while traveling alone?

- **Section 3: Digital Safety Solutions & Technology Preferences**

- What digital tools or safety apps do you currently use while traveling?
- What features do you currently find most useful in travel safety apps, and what additional features would you like to see?
- How do you evaluate the reliability of safety recommendations from AI or digital platforms?
- Would real-time, crowdsourced safety ratings influence your travel decisions? Why or why not?

- **Section 4: Community & Social Support in Travel Safety**

- How do online communities and peer networks influence your travel safety decisions?
- How would having access to verified local guides impact your sense of safety and travel confidence?
- What are your thoughts on buddy systems for solo travellers? Have you used one before?

- **Section 5: User Needs & Suggestions for Safety Solutions**

- If you could design an ideal solo travel safety tool, what features would it include?
- Based on your experiences, what improvements would you suggest for digital tools designed to enhance solo travel safety?
- What kind of support or solutions would help you feel more secure when traveling alone?

A.3 Usability Testing

Steps Followed:

- (1) Briefing users on test objectives.
- (2) Assigning them realistic scenarios (aligned with actual app features).
- (3) Observing task completion and noting time, pain points, likings, suggestions.
- (4) Collecting post-task feedback via discussion and a 1–5 scale rating.
- (5) Aggregating results to form usability metrics (Learnability, Efficiency, etc.).

Use Cases Tested:

- **Sign-in & Onboarding:** Evaluate ease of account creation, identity verification, and initial app comprehension.
- **Trusted Contacts:** Test how easily users can add, manage, and view trusted contacts.

- **Post-Trip Feedback:** Check if users understand how and where to give feedback (stars, written, image).
- **Search & Explore:** Assess discoverability, filtering, and clarity of safety scores & destination info.
- **Buddy System:** Evaluate how users find, connect, and communicate with verified buddies.
- **Emergency & SOS:** Test how quickly and accurately users can activate SOS and feel confident doing it.
- **Real-Time Navigation:** Examine usability of safe route selection, alerts, live location sharing & reporting.

Usability Testing Matrix:

Use Case	Test Objective	Test Scenario
1. Sign-in & Onboarding	Evaluate ease of account creation, identity verification, and initial app comprehension	"Create a new account, verify it with ID and camera, and reach the app's homepage, add emergency info"
2. Trusted Contacts	Test how easily users can add, manage, and view trusted contacts	"Add your sister as a trusted contact, then delete an existing one and view a contact's details."
3. Post-Trip Feedback	Check if users understand how and where to give feedback (stars, written, image)	"Submit feedback for your recent trip to barcelona, add a photo, and report a minor incident."
4. Search & Explore	Assess discoverability, filtering, and clarity of safety scores & destination info	"Search for a nearby nature spot in mountains and look at it ratings"
5. Buddy System	Evaluate how users find, connect, and communicate with verified buddies	"Find a verified female local guide nearby, send her a request, and start a chat if she accepts."
6. Emergency & SOS	Test how quickly and accurately users can activate SOS and feel confident doing it	"You feel unsafe. Trigger SOS and describe what happens. (Mock simulation)"
7. Real-Time Navigation	Examine usability of safe route selection, alerts, live location sharing & reporting	"Navigate to 'Nidhari Reservoir,' choose the safest route, and share your live location."

Figma Tool: The tool Figma was used for designing wireframes, storyboarding, and prototyping the app.